

Exploratory Data Analysis on the Titanic Dataset

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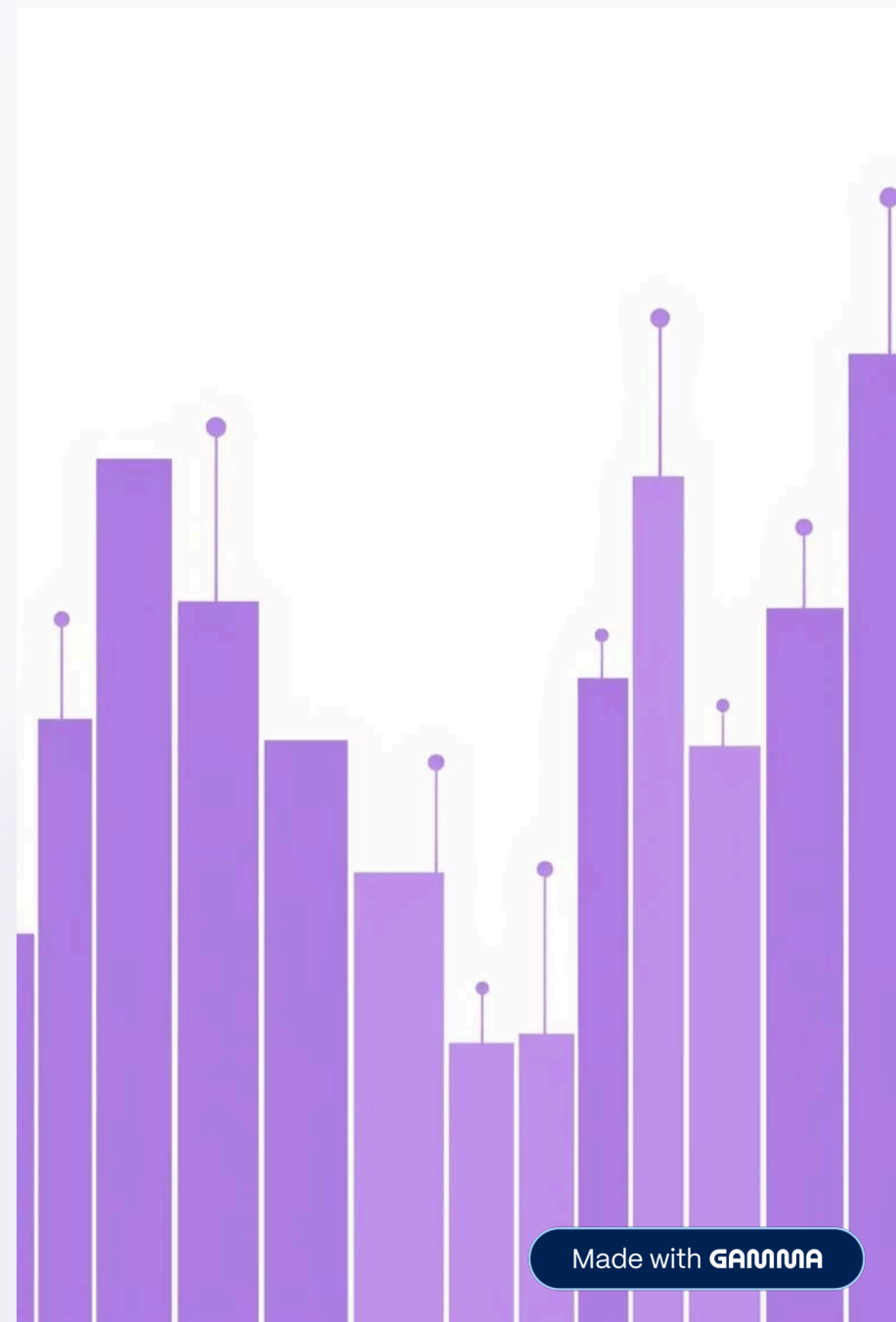
Introduction to the Titanic Dataset

This project focuses on performing Exploratory Data Analysis (EDA) on the Titanic dataset. We will use Python libraries like Pandas, Seaborn, and Matplotlib to visualize and identify survival patterns among passengers.

Dataset Overview

The dataset contains 891 entries and 12 columns. Key columns include 'Survived', 'Class', 'Gender', and 'Age'.

Missing values are present in 'Age' (177), 'Cabin' (687), and 'Embarked' (2).



Objectives of EDA

1 Understand Data Structure

Examine dataset shape, info, and missing values.

2 Visualize Survival Patterns

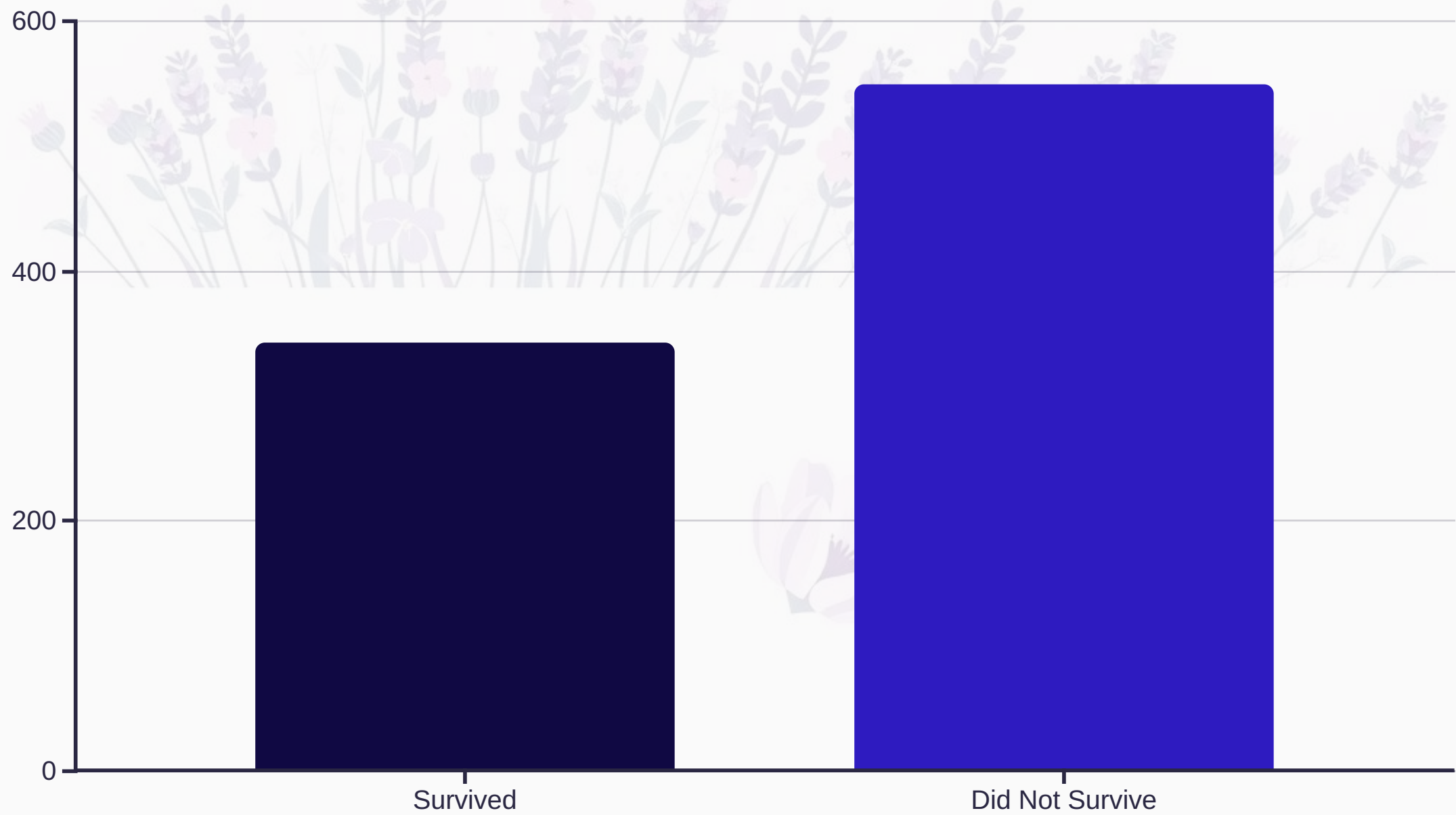
Identify trends in survival based on various factors.

3 Extract Key Insights

Derive meaningful conclusions from the visualizations.

Survival Count on the Titanic

This chart illustrates the stark reality of survival on the Titanic, showcasing the distribution of passengers who survived versus those who did not.

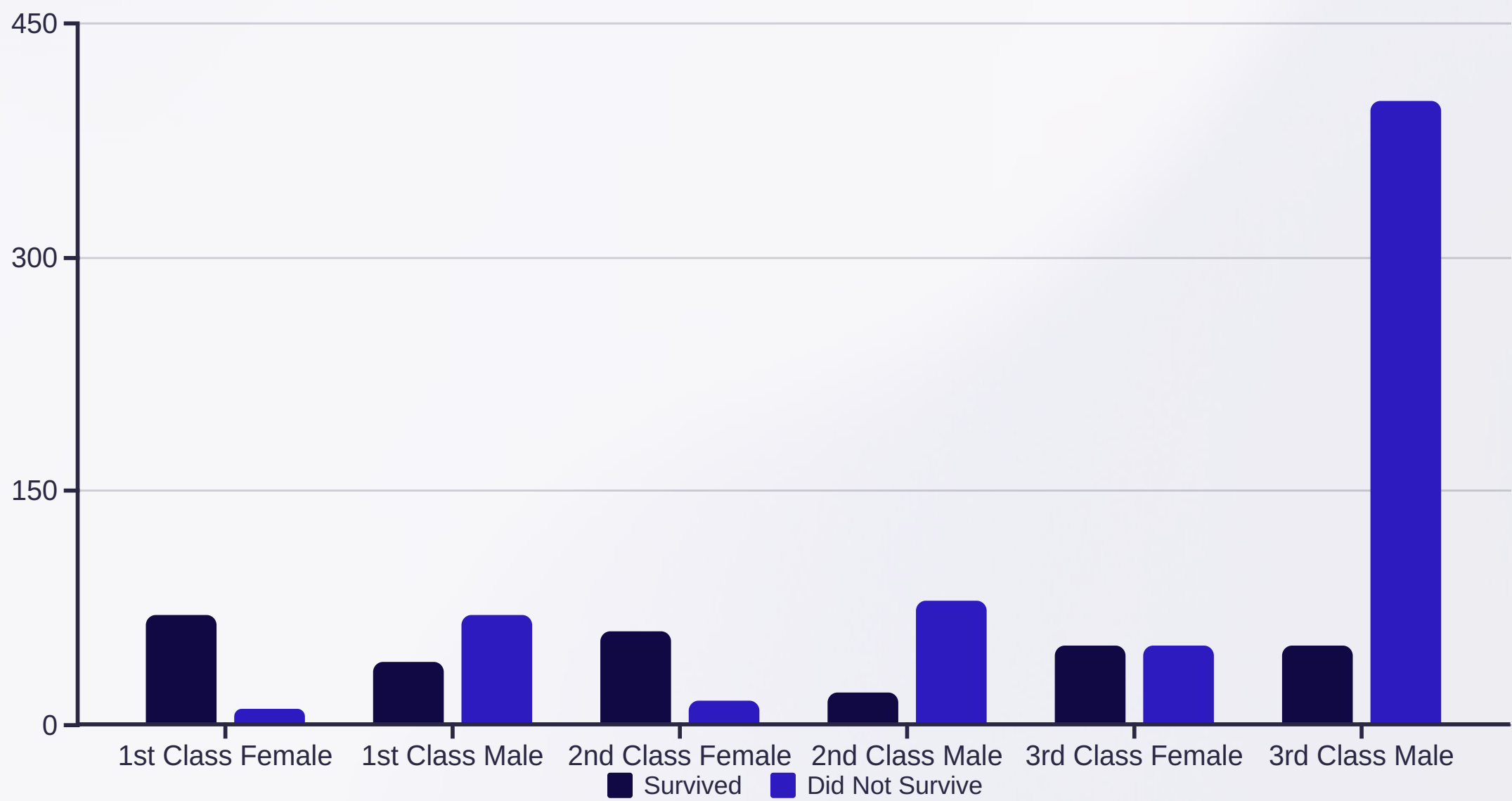


As depicted, a significant majority of passengers did not survive the disaster.



Survived Passengers by Gender and Class

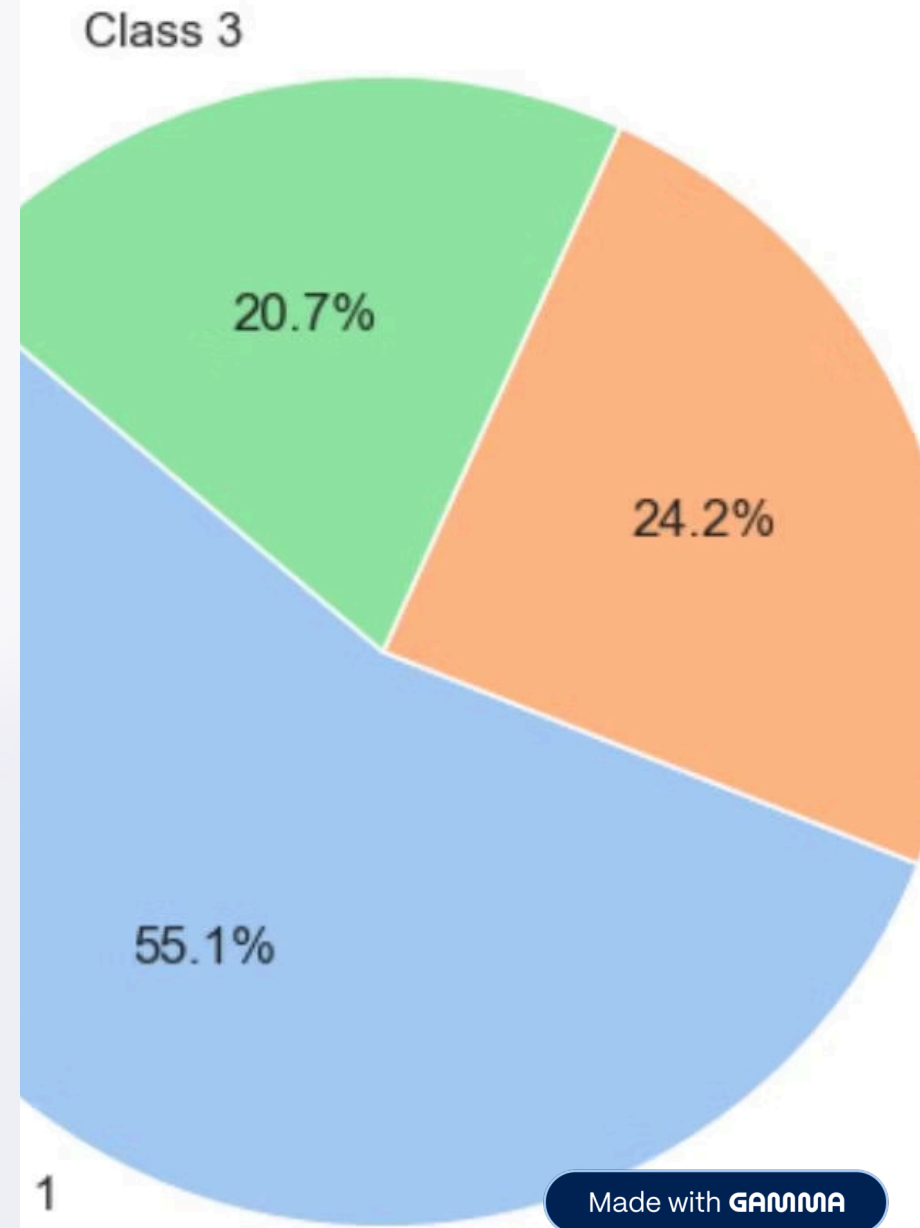
This chart vividly illustrates survival rates on the Titanic, categorized by passenger class and gender. It highlights significant disparities in who survived the disaster.



As observed, females, particularly those in higher classes, had a significantly greater chance of survival compared to males across all classes.

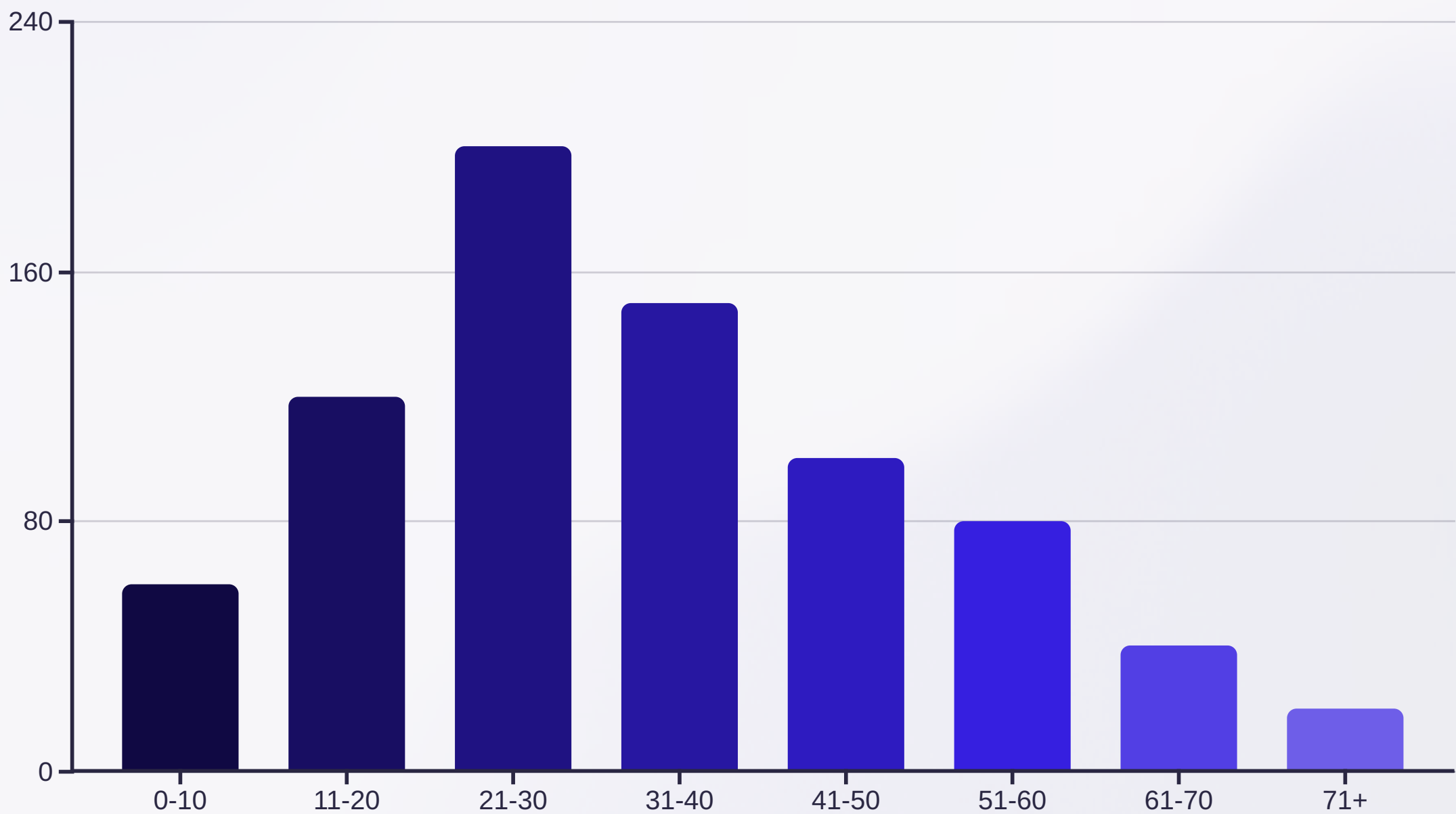
Passenger Class Distribution

The pie chart illustrates the proportion of passengers in each class, with Class 1 having the largest share.



Age Distribution of Passengers

This bar chart illustrates the distribution of passengers across different age groups on the Titanic, highlighting the predominant age ranges onboard.



As observed, there was a higher concentration of passengers in the younger to middle age groups, particularly between 21 and 40 years old.

Key Insights and Conclusion

- **Survival Bias:** Females and higher-class passengers had better survival chances.
- **Age Impact:** Age distribution provides context for passenger demographics.
- **Data Gaps:** Significant missing data in 'Cabin' and 'Age' require attention for further analysis.

EDA on the Titanic dataset reveals crucial patterns, guiding deeper predictive modeling.